

Lóránt Nagy, PhD

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Professional Experience

Eötvös Loránd University, ELTE AI Research Group

2023–Present

Research Assistant

- Research deep-learning methods for statistical inference in stochastic-process and long-memory time-series models.
- Conducted an interpretability study on neural networks applied to time-series data: investigated which characteristics of the input drive the networks' inference.
- Lead technical development of reinforcement-learning methods for automated trading in stylized market environments.

Alfréd Rényi Institute of Mathematics

2019–Present

Research Fellow

- Investigate alternative stochastic dynamics for diffusion-based generative models.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory

2024

ML Researcher

- Collaborated with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models, with a focus on predictive reliability.

Sysler Kft.

2023

Research Consultant

- Prototyped a Bayesian late-fusion framework combining outputs from independent medical-diagnostic models to infer disease progression from empirical patient trajectories.

Morgan Stanley

2017

Interest Rate Modelling Intern

- Reviewed financial-product documentation, checking model assumptions, calculations, pricing results, and internal consistency.
- Reverse-engineered parts of a legacy pricing system written in A+ to understand existing model and pricing logic.

Education

Central European University

2018–2022

PhD in Mathematics

Budapest University of Technology and Economics

2015–2018

MSc in Mathematics

Technical Skills

Programming and development tools: Python, PyTorch, scikit-learn, NumPy, SciPy, Git, Docker, GitHub Actions, Wolfram Mathematica, TensorBoard, Optuna

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, mathematical finance, statistical learning, time-series forecasting, information theory

Cloud and deployment: Azure Static Web Apps, Azure Container Apps, containerized deployment, basic CI/CD workflows

Machine learning and deep learning: generative modelling, neural networks, variational inference, reinforcement learning, late fusion, Bayesian filtering, tree-based methods, dimensionality reduction, clustering

Selected Publications

2026 Rásonyi, M. and Nagy, L.

Optimal investment with transient price impact and stochastic spread.

Preprint, arXiv:2511.12093.

2024 Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L.

Deep learning the Hurst parameter of linear fractional processes and assessing its reliability.

Quality and Reliability Engineering International, 40, 4228–4246.

2022 Guasoni, P., Nagy, L., Rásonyi, M.

Young, timid, and risk takers.

Mathematical Finance. 31: 1332–1356.

2021 Rásonyi, M. and Nagy, L.

Optimal long-term investment in illiquid markets when prices have negative memory.

Electronic Communications in Probability, 26.

Selected Talks

2026 **Bologna** XIII Bachelier World Congress

On the utility problem in a market where price impact is transient

2025 **Vienna** Congress on Mathematical Finance, TU Wien

Non-convex optimal investment with transient price impact

2024 **Szeged** Conference on Analysis and Applied Mathematics, University of Szeged

Optimal execution in superlinearly mean-reverting environments

2023 **Budapest** ELTE–Rényi AI Workshop, Eötvös Loránd University

Risk-averse trading

2022 **Ann Arbor** Financial Mathematics Seminar, University of Michigan

Young, Timid, and Risk Takers

Additional Experience and Distinctions

2025 Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.

2022 Award for Advanced Doctoral Students.

2019 Instructor, *Fundamentals of Stochastic Analysis*, Central European University.

2016 Second Place, TDK competition, *Examples in the theory of continuous trading with friction.*